



Harry Locomotive Project

How the Collision Prevention System Works

A simple guide for Piotr | Prepared by Ahmed Ali, CEO — Datix AI | May 2026

What Does This System Do?

The ESP32 microcontroller is the brain sitting between your two trains. Its job is simple — make sure the inner train and outer train never collide inside the shared section of track. It watches three sensors 24 hours a day and automatically stops the right train at the right moment, then lets it go again when the coast is clear.

Your Two Trains

OUTER LOOP TRAIN (Blue)	INNER LOOP TRAIN
Runs continuously on the outer loop Controlled by Bluetooth (BLE) ESP32 sends speed/stop commands wirelessly Starts automatically when system boots	Stays PARKED by default — engine off Controlled by relay cutting track power Press X in serial monitor to run ONE loop Stops automatically at parking spot after loop

The Three Sensors — What They Watch

SENSOR A GPIO 16	Inner loop entry. Only the inner train can pass here. When it fires — outer BLE train receives an immediate STOP command and waits outside the shared section.
SENSOR B GPIO 15	Outer loop entry. Only the outer train can pass here. When it fires — the relay cuts power to the inner train track so it cannot enter the shared section at the same time.
SENSOR C GPIO 17	Shared section exit. Any train passing here signals it has left the shared section. The system then waits 2 seconds for safety, then resumes the stopped train.

What Happens Step by Step

1 Inner Train Enters	You press X. Inner train gets track power and starts running. Sensor A detects it entering the shared section.
2	ESP32 immediately sends Bluetooth STOP to the outer train. It stops safely before the shared section. STOP is repeated every 0.5 seconds so it cannot creep forward.

Outer Train Waits

3 Inner Train Exits Inner train passes Sensor C. It has 2 seconds to coast to its parking spot, then the relay cuts power. It stops exactly at the parking position.

4 Outer Train Resumes After a 2 second safety gap, outer train slowly resumes, then ramps up to full speed. Zone is clear — everything resets.

Keyboard Commands (in Serial Monitor)

Key	What it does
x	Start inner train — runs ONE full loop, then parks automatically
z	Emergency stop — cuts inner train power immediately
s	Stop the outer BLE train
f	Set outer train to go Forward
1-7	Set outer train speed (1=slow, 7=fast)
+ / -	Increase or decrease outer train speed by one step
p	Manually pulse the track switch for 1 second
i	Show full system status — sensor states, errors, timers
?	Show the full command menu

The Two Relays — What They Control

Relay 1 — Track Power (GPIO 18) Controls whether the inner train has electricity flowing to its track. ON by default — inner train has power. Turns OFF only during collision prevention to stop the inner train.	Relay 2 — Track Switch (GPIO 19) Resets the track switch position so the outer train takes the correct track after the shared section. Only pulses for exactly 1 second. Only fires when the OUTER train has exited — never for the inner train.
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Built-in Safety Features

Relay boot fix	Both relays start OFF as the very first action on power-up. Prevents accidental triggering during start-up.
Repeated STOP signal	BLE STOP is sent every 0.5 seconds while zone is locked — so the outer train cannot creep forward even if one packet is missed.

30-second timeout	If Sensor C never fires (train stuck or derailed), system automatically resumes after 30 seconds to prevent permanent lockup.
Sensor health check	Every sensor is checked continuously. If a sensor is stuck or misaligned, a clear warning is shown in the serial monitor.
Hardware watchdog	If the ESP32 software ever freezes, a hardware timer automatically resets it within 60 seconds. Restart count is logged.
BLE auto-reconnect	If Bluetooth drops, ESP32 reconnects automatically every 5 seconds. If reconnected during a locked zone, re-sends STOP immediately.

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